Types of Software Used in the Project

Frontend Software

Purpose: To create the user interface and manage user interactions.

HTML (HyperText Markup Language): Used to design the structure of web pages−where farmers can input details such as crop type, soil type, and water level.

CSS (Cascading Style Sheets): Used for styling the web pages, including colors, fonts, spacing, and layouts, making the interface clear and easy to use.

JavaScript: Adds interactivity to the website. It enables dynamic updates (e.g., showing irrigation suggestions instantly) without reloading the page.

Backend Software

Purpose: To handle logic, process data, and connect the frontend with the database.

Node.js: Used for processing data received from the frontend. It runs the logic to generate irrigation suggestions based on crop, soil, and water level, and interacts with the database to store or retrieve information.

Database Software

Purpose: To store, manage, and retrieve data efficiently.

MongoDB: A NoSQL database used to store various data such as crop types, soil properties, water levels, and irrigation recommendations. It supports fast data access and flexibility in managing different data types.

Cloud and Hosting Software

Purpose: To host the project online and store data securely for global access.

Amazon Web Services (AWS): Provides cloud infrastructure to host the web application and manage data storage securely.

- **Amazon S3:** Used to store images, videos, or other media files.
- AWS Database Hosting: Ensures accessibility of the application for farmers anywhere.

Development Software

purpose: To write, edit, and manage code efficiently.

Visual Studio Code (VS Code): The primary development environment used for building the entire project. It provides features like syntax highlighting, debugging, and integrated terminal support, making development faster and more efficient.

Conclusion

This project combines different types of software to create a complete and efficient irrigation management system:

- **Frontend software** for design and interactivity,
- **Backend software** for logic and data processing,
- **Database software** for data management,
- **Cloud software** for hosting and accessibility, and
- **Development software** (VS Code) for coding and building the project.